

# MANAV GOEL

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## Summary

Final-year CS student who builds and ships LLM systems end to end, from retrieval and agent design through FastAPI backends, evaluation and production deployment. Two AI internships and a steady run of self-built agents and RAG products, with a habit of measuring what gets built (recall/MRR baselines, grading, citation checks) rather than assuming it works.

## Education

**University School of Information, Communication and Technology, GGSIPU**

*B.Tech in Computer Science Engineering | CGPA: 8.1/10*

**2023 – Present**

*New Delhi, India*

## Experience

**AI Engineer Intern**

**Jun 2026 – Aug 2026**

**IntelGrader**

*Remote*

- Developed a RAG-based AI assessment pipeline using Python, Gemini, LangChain and ChromaDB, processing 50+ educational documents with hybrid semantic search, BM25 reranking and LLM-based query routing to achieve 95%+ chapter-classification accuracy.
- Automated answer generation and evaluation for 1200+ exam questions, achieving up to 92% benchmark accuracy and reducing manual content-review effort through structured retrieval, evaluation and grading workflows.
- Engineered a multi-agent AI pipeline turning Class 8–12 STEM content into interactive educational webpages across 12 visual families (LLM orchestration, Node.js, JavaScript, HTML/CSS/SVG, browser automation), with automated validation and self-repair across 25+ generated experiences that cut recurring generation and rendering failures.

**AI/ML Intern**

**Dec 2025 – Feb 2026**

**Hyrte**

*New Delhi, India*

- Redesigned LLM prompt orchestration using LangChain across the interview pipeline, improving conversational coherence and reducing repetitive system responses.
- Engineered a dual-layer evaluation system (heuristic + evidence-based) with adaptive difficulty progression and priority-based topic ordering, improving assessment accuracy across technical domains.
- Optimized prompt structure and rolling context windows, reducing LLM token usage by 42% and improving response latency within a FastAPI backend and validated APIs using Postman.

## Projects

**InsightGraph – Autonomous AI Intelligence Pipeline** | *Python, FastAPI, LangGraph, LangSmith, pgvector* | [GitHub](#) | [Live](#)

- Architected a 6-agent LangGraph pipeline with a two-stage hallucination firewall: a semantic validator that filters irrelevant signals pre-ingestion and a final evaluator enforcing word-count and diversity constraints, with dynamic conditional retry routing to recover from failed generations.
- Implemented memory-augmented RAG with pgvector (384-dim embeddings), enabling agents to query historical briefings for longitudinal trend context; resolved a production DNS-blocking issue on Render by migrating embedding inference to HuggingFace's InferenceClient SDK.
- Cut compute costs via tiered LLM routing (8B model for parallel relevance filtering, 70B for deep analysis/composition) and automated daily delivery through a GitHub Actions cron job engineered to keep a free-tier Render backend awake through the full pipeline run.

**Groundwork – Cited Engineering Q&A Agent** | *Python, FastAPI, LangGraph, PostgreSQL, Next.js, Docker* | [GitHub](#) | [Live](#)

- Built a LangGraph agent that answers questions about engineering work across GitHub, Jira and Slack with a citation for every claim: exact ones ("last commit by X") route to typed SQL in under 25 ms with no model call, the rest to hybrid full-text + pgvector retrieval fused by reciprocal rank fusion (recall@8 0.807, MRR 0.941).
- Enforced answer integrity with a model-based sufficiency grader (0.950 accuracy) driving a bounded CRAG corrective loop, per-marker citation validation, and an entailment check that downgrades and discloses any claim its cited passage does not state (recall 0.909).
- Deployed full-stack to production (Next.js on Vercel; FastAPI + Postgres on EC2 behind Caddy/TLS), cutting the RAG turn from 67.9 s to 1.8 s, and gated releases on 340+ tests plus eval suites, including a generalization suite that scored 8/8 on an unseen project.

## Technical Skills

- Languages:** Python, TypeScript, SQL, C++
- AI & ML:** LangGraph, LangChain, RAG, Hybrid Retrieval, Reranking (RRF), Prompt Engineering, Generative AI
- Evaluation:** Eval-harness design, DeepEval, LangSmith, recall@k / MRR, citation & entailment checks, pytest
- Backend & Data:** FastAPI, REST APIs, SQLAlchemy, Alembic, PostgreSQL, pgvector, ChromaDB, Next.js, React
- Models & Infra:** Ollama, Gemini, Groq, HuggingFace, Docker, AWS EC2, GitHub Actions, Vercel, Caddy

## Positions of Responsibility

**Lead, ACM ICPC Club** | *USICT, GGSIPU*

**2024 – 2026**

- Organized coding contests and DSA workshops, driving competitive programming engagement across campus.
- Mentored 150+ junior students in algorithms and problem-solving, boosting participation in national coding competitions.

## Achievements

- Solved 700+ DSA problems across LeetCode, GeeksforGeeks and CodeStudio ([Profile](#)); completed Coding Ninja Slayground 2.0, earning the Level 3 Conqueror Certificate ([Certificate](#)).